

From local to the whole: understanding construction projects as multiteam systems

From local to
the whole

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Jianyao Jia

School of Economics and Management, Tongji University, Shanghai, China

Yang Li and Xueqing Gan

Tongji University, Shanghai, China, and

Guofeng Ma

School of Economics and Management, Tongji University, Shanghai, China

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Abstract

Purpose – In today's digital and post-pandemic era, construction teams can span their boundary to obtain important resources and support in computer-mediated ways. However, the benefits of computer-mediated team boundary spanning (TBS) are mostly assumed. Empirical evidence for these benefits is in lack. Thus, this study attempts to investigate the influence of computer-mediated (instant messaging in this study) TBS on construction project performance and the underlying mechanism.

Design/methodology/approach – From the multiteam system (MTS) perspective, this study employs the input-process-outcome (IPO) framework and coordination theory to explore the influence of instant messaging-based team boundary spanning (IMTBS) on construction project coordination processes and project performance. A theoretical model is built and validated with a survey involving 206 construction projects.

Findings – Results show that IMTBS positively affects knowledge integration and project coordination, but negatively affects project commitment. The positive link between IMTBS and project performance is attributable to knowledge integration and project coordination. Further, project commitment is also found to negatively mediate the relationship between IMTBS and construction project performance.

Originality/value – To the best of the authors' knowledge, this study is the first to empirically investigate the influence of TBS on construction project performance. As such, this study contributes to the literature on TBS in the construction field. In addition, the findings of this study unveil the double-edged characteristic of IMTBS, which should attract managers' attention. This also indicates that the form of TBS is a significant determinant of its effectiveness, which should attract scholars' attention in the future.

Keywords Construction project, Multiteam system, Team boundary spanning, Instant messaging, Coordination theory

Paper type Research paper

1. Introduction

A construction project can be viewed as a multiteam system (MTS), which consists of a set of teams (also component teams) from different firms such as design teams and construction teams (Garcia and Mollaoglu, 2020a, b; Ma *et al.*, 2020). Due to various interdependencies, a component team has to span its boundary to interact with other teams, so as to achieve its own goal. This process is referred to as team boundary spanning (TBS). TBS is defined as actions a team undertakes to reach out into its environment to obtain important resources and support (Faraj and Yan, 2009). Example actions include updating and seeking feedback from other teams, and communicating action plans and delivery deadlines with other teams (Marrone, 2010).

Although previous studies have concluded that TBS generally contributes to team performance (Carbonell and Rodríguez Escudero, 2019; Marrone *et al.*, 2007), mixed results



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are found in the literature concerning the relationship between TBS and MTS (system-level) performance. [Gopal and Gosain \(2010\)](#) suggested that TBS between the vendor and the client could improve project performance in outsourcing and software projects. However, [Davison *et al.* \(2012\)](#) indicated that TBS between component teams weakens MTS performance in an experimental study. In multiteam R&D projects, [Hoegl *et al.* \(2004\)](#) found that TBS has differential effects on different performance indicators. [Luciano *et al.* \(2018\)](#) and [Zaccaro *et al.* \(2020\)](#) believed that the form of MTS is the main reason for the above inconsistency. Thus, both of them appealed for nuanced investigation in specific MTS contexts. As a result, previous findings cannot be extended to the construction project context. Besides, to the best of our knowledge, previous research has paid little attention to the influence of TBS on construction project performance. Therefore, it is relevant to empirically examine the influence of TBS on MTS performance in the construction project context.

The temporal and spatial distribution of construction work induces great demands on computer-mediated or digital tools for project organizing ([Jia *et al.*, in press](#); [Jia *et al.*, 2021a](#); [Ma *et al.*, 2021](#)). In today's post-pandemic era, such demands are further expanded ([Leonardi and Treem, 2020](#); [Margherita *et al.*, in press](#)). Consequently, as a kind of daily organizing practice, TBS is increasingly mediated by computer-mediated technologies ([Alin *et al.*, 2013](#)). Despite tangible benefits, computer-mediated TBS could also suffer from corresponding drawbacks. Recent studies found that power holders who control important resources (i.e. important actors) feel more responsible for others in face-to-face environments ([Scholl *et al.*, 2020](#)), thus decreasing the effectiveness of TBS. In addition, in a collective online environment, those who are not involved in interactions can feel left out, and this can induce unintended consequences ([Majchrzak *et al.*, 2004](#)). These scattered but indicative cues suggest that the effectiveness of computer-mediated TBS is doubtful, especially in construction projects featured by large MTS ([Kou, 2021](#)). In this way, our attention to computer-mediated TBS is relevant and timely.

As a flexible digital tool, instant messaging enables not only interpersonal contact but also group communication, and facilitates not only text contact but also video, image and audio communication. Instant messaging has been widely adopted for both formal and informal organizing activities in the construction industry ([Ma *et al.*, 2021](#); [Pivec and Maček, 2019](#); [Sheer and Rice, 2017](#); [Zhang *et al.*, 2018](#)). In this way, this study specifically focuses on instant messaging-based team boundary spanning (IMTBS) and explores its effect on construction project performance. IMTBS encompasses TBS activities mentioned in the first paragraph, but in a flexible way, which means removing time and space limits, enabling dynamic, interactive and collective boundary-spanning activities ([Jia *et al.*, in press](#); [Sheer and Rice, 2017](#)). Meanwhile, the above-stated drawbacks of computer-mediated TBS apply to IMTBS.

To explore the underlying mechanism between IMTBS and construction project performance, this study employs and combines the input-process-outcome (IPO) framework and coordination theory ([Crowston, 1997](#); [Marks *et al.*, 2001](#)). IPO framework is employed for its effectiveness in explaining system functioning process and mechanism. Coordination theory is adopted for the following several reasons. First, in MTSS, system-level coordination is the core organizing process and is fundamental to MTS functioning and performance ([Davison *et al.*, 2012](#); [Luciano *et al.*, 2018](#)). Second, TBS is the activity interfacing component teams, thus presumably affecting MTS coordination ([Kou, 2021](#); [Zaccaro *et al.*, 2020](#)). Third, coordination exhibits in a variety of forms, including task coordination, knowledge coordination and relational coordination ([Bolton *et al.*, 2021](#); [Crowston, 1997](#)). The differential mediating effects deserve our careful attention. Fourth, and finally, previous studies have not explored the linking mechanism between IMTBS and MTS performance from a coordination perspective, especially in construction projects.

This study attempts to make the following several contributions. First, by empirically investigating the influence of IMTBS on construction project performance, this study enriches the literature on TBS in the construction field. Second, this study explores the

mechanism linking IMTBS and construction project performance. The results indicate significantly different effects of the three forms of coordination between IMTBS and project performance. In this way, these findings provide us with a nuanced understanding of the relationship between IMTBS and project performance. Third, this study reveals both the positive and negative facets of IMTBS in a theoretical model, which explains the inconsistency in the effect of TBS on MTS performance to some degree. Thus, this study also contributes to the MTS literature.

2. Literature review

2.1 Team boundary spanning

TBS becomes increasingly important as organizational work teams must couple with the increased complexity of work tasks, flatter organizational structures and changing environmental conditions (Marrone, 2010). TBS encompasses a set of different activities, mainly including ambassador, scouting and task coordination, through which teams actively manage interactions with their embedding external environment (Ancona and Caldwell, 1992; Marrone, 2010). In construction projects, although the external environment may refer to teams nested within and actors outside of the project (Marrone *et al.*, 2007), this study focuses on TBS within the project.

In the construction management field, boundary spanning has attracted scholars' attention, which mainly includes the identification of boundary-spanning practices (Sackey and Akotia, 2017; Stjerne *et al.*, 2019) and individual boundary-spanning behaviors (Di Marco *et al.*, 2010; Lee and Sawang, 2016). To deal with temporal tensions among project parties, Stjerne *et al.* (2019) adopted a practice-based theory to identify interorganizational boundary-spanning practices. Framing, synchronizing and hyping are three practices that potentially resolve the temporal tensions in construction projects. Lee and Sawang (2016) explored the antecedents of team members' boundary-spanning behavior. They found that group attachment anxiety positively affects boundary-spanning behavior, whereas group attachment avoidance makes a negative effect. On the one hand, these studies mostly adopted an interorganizational or individual perspective (Garcia and Mollaoglu, 2020a, 2020b). Yet, practically, different parties organize their work in a team way. Simply, the team-based perspective reflects the actual organizing process of construction activities, which has been ignored in the literature. On the other hand, previous studies focused more on antecedents of boundary-spanning behaviors but ignored the effects induced by team boundary spanning, thus leaving a gap for this study.

2.2 Coordination theory

Coordination is defined by Malone and Crowston as "the act of managing interdependencies between activities performed to achieve a goal" (Malone and Crowston, 1990). From a macro perspective, coordination theory concerns how to subdivide goals or tasks, how to allocate resources and how to integrate efforts contributed by different actors (Crowston, 1997; Malone and Crowston, 1990).

In construction projects, coordination aims at integrating and aligning the actions, knowledge and objectives of interdependent teams in order to attain common goals (Rico *et al.*, 2008). In this way, three forms of coordination are formulated: task/project coordination, knowledge integration and project commitment. Project coordination involves task interdependencies and is defined as managing dependencies between project activities (Jia *et al.*, 2021b). Knowledge integration concerns knowledge interdependencies and refers to the process to synthesize different types of knowledge (Okhuysen and Eisenhardt, 2002). Project commitment refers to the acceptance of and the strong belief in the goals and values of the

project (Hoegl *et al.*, 2004), which relates to objective interdependencies and relational coordination (Bolton *et al.*, 2021).

TBS is regarded as the mechanism linking the local team and other component teams (Kou, 2021). This process inevitably involves the management of interdependencies between different teams. From a coordination structure perspective, previous research has argued that boundary-spanner roles are critical to coordinating work across different parties or teams (Bolton *et al.*, 2021). Crowston (1997) suggested coordination mechanisms are primarily information-processing activities, and thus information technology contributes a lot. Taken together, this study considers IMTBS as a kind of potential coordination mechanism and therefore explores its efficacy in promoting coordination processes.

Figure 1 depicts the research model of this study.

3. Hypothesis development

3.1 Effects of instant messaging-based team boundary spanning

Instant messaging enables one-to-one, one-to-many and many-to-many communication as well as the transfer of different types of information (Chatterjee *et al.*, 2017; Zhang *et al.*, 2018), by which each team can span its boundary. Example team boundary-spanning actions include influencing important actors of other teams through one-to-one communication and presenting their activities to other teams through one-to-many communication. When an individual team tries to span boundary by instant messaging, it intends to seek support or solicit information and resources from other teams (Faraj and Yan, 2009). When the individual team intends to acquire resources from other teams, other teams commonly need to seek support from networked teams to fulfill the team’s requirements. Such a ripple process, which originated from an individual team’s boundary spanning, enables the construction project to function as a whole.

We suggest that the ripple process transforms an individual team’s boundary spanning into joint efforts for the project. As previous research indicates, building online communities or groups is increasingly prevalent for TBS in project management (Pivec and Maček, 2019). When an individual team encounters troubling problems, it can seek advice and support from other teams by describing its problems in their common instant messaging-based groups (IMGs). In this way, the affordances of meta-voicing and triggered attending enabled by IMGs facilitate the contribution of other teams’ expertise to jointly solve their problems (Majchrzak *et al.*, 2013). In this process, the individual team can obtain creative ideas and schemes, and other teams can also gain insights into the project. The communication visibility affordance of instant messaging enables TBS to occur in an interactive, dynamic and timely manner, thus contributing to knowledge creation and integration (Jia *et al.*, in press). Ratcheva (2009) has also indicated the potential of IMTBS for knowledge integration in multidisciplinary project teams. Thus, the following hypothesis is proposed:

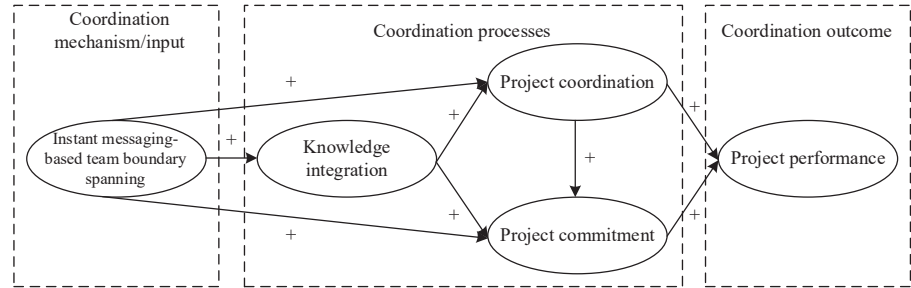


Figure 1.
Conceptual model

H1. IMTBS is positively related to knowledge integration.

Besides, the individual team can inform other teams of its activities in their common IMGs to avoid work duplication for project coordination. TBS commonly means resource acquisition (Marrone, 2010); thus, in a passive way, the construction project has to coordinate the limited resources. In practice, poor task coordination is mainly attributed to poor communication among component teams in construction projects (Du *et al.*, 2019). The flexibility of instant messaging lies in its potential for both distributed and centralized communication structures, which greatly satisfies extensive communication demands in construction projects (Kou, 2021). In addition, IMTBS removes both time and space barriers to construction coordination. Therefore, we suggest the following hypothesis:

H2. IMTBS is positively related to project coordination.

A positive relationship between IMTBS and project commitment is also expected. Bolton *et al.* (2021) suggested that frequent and timely communication is the foundation of relational coordination. On the one hand, frequent interactions help to learn other teams' tasks and see the whole system structure and processes (Kou, 2021). On the other hand, timely feedback fosters a sense of recognition, which contributes to collective identity and common goals (Hasan *et al.*, 2019). Of note, it is instant messaging that makes frequent and timely boundary spanning possible. Additionally, IMGs function as shared spaces, which facilitate communication visibility (Jia *et al.*, in press). Communication visibility increases process transparency, fosters mutual trust and cultivates a shared mental model, which all contribute to the acceptance of common goals and values (Leonardi, 2014, 2015). Informed by these arguments, we put forth the following hypothesis:

H3. IMTBS is positively related to project commitment.

3.2 Effects of knowledge integration

In the construction project context, knowledge integration is a crucial process of knowledge management. It broadly refers to the assimilation and consolidation of each team's specialized knowledge within the project (Mehta and Mehta, 2018; Yang *et al.*, 2020). Knowledge integration is commonly embedded in knowledge conversation (i.e. internal project meetings), in which teams express opinions and ideas based on their respective knowledge structures and expertise. Of note, communication and discussion among teams aim to deal with specific tasks and problems. In this way, the knowledge integration process deepens mutual understanding of other teams' task requirements and knowledge structures, which will promote project coordination (Kanawattanachai and Yoo, 2013). In the construction management field, the positive effects of knowledge processes on project coordination have been empirically confirmed (Ma *et al.*, 2021; Wen and Qiang, 2016; knowledge sharing in their studies). Consistent with previous literature, the following hypothesis is proposed:

H4. Knowledge integration is positively related to project coordination.

Knowledge integration is also expected to improve project commitment. When the construction project encounters challenging issues, only when each team pools its expertise, will these project-related problems be solved. In such a process, each team will feel valued by others, and thus all teams are likely to feel responsible not only for their team performance but also for the overall outcome of the project (Buvik and Tvedt, 2017). Besides, knowledge integration is commonly embedded in the process of problem-solving. Successful problem-solving with joint efforts strengthens the strong belief in the goals of the project as well as the desire to maintain membership. In public sectors, Razzaq *et al.* (2019) have verified

the positive effect of knowledge management on organizational commitment. In this situation, combined with the above argument, we propose the following hypothesis:

H5. Knowledge integration is positively related to project commitment.

3.3 Effects of project coordination

We expect a positive relationship between project coordination and project commitment in construction projects and provide the following explanation. Project coordination means that each team has to rely on what other teams are going to do (Palm and Lindahl, 2015). The project work done by each party is closely harmonized in a well-coordinated project, thus avoiding duplication of efforts (Siebdrat *et al.*, 2014). In this way, each team will perceive a common fate and focus on the common project (Hoegl *et al.*, 2004). From the perspective of psychology, interdependent relationships also reflect expectations about other teams' upcoming actions (Michael *et al.*, 2016). Social pressure originating from such expectations compels each party to try to avoid disappointing the others' expectations and to remain engaged until the project is completed (Michael *et al.*, 2016). Furthermore, a recent study by Le Dain *et al.* (2020) has also demonstrated a positive effect of project coordination on project commitment in the new product development project context. Consequently, we put forward the following hypothesis:

H6. Project coordination is positively related to project commitment.

The relevance of coordination has also been emphasized in various project contexts, especially in multiteam projects (Kujala *et al.*, 2021). Project coordination functions a central role in effective MTS functioning as dependencies among teams should be well managed (Hossain, 2009; Luciano *et al.*, 2018; Zaccaro *et al.*, 2020). Project coordination means that each team has to rely on what other parties are going to do (Palm and Lindahl, 2015). The project work done by each team is closely harmonized in a well-coordinated project, thus avoiding duplication of efforts (Siebdrat *et al.*, 2014). Further, project coordination has been identified as a key process factor of construction project performance (Lindhard and Larsen, 2016). Project coordination ensures the project as a whole to function (Garcia and Mollaoglu, 2020a, b). If teams within the project perform their tasks separately, the final project deliverable tends not to function, that is, project failure. Besides, project coordination guarantees a harmonious work way, thus avoiding duplication of efforts (Siebdrat *et al.*, 2014). Therefore, the following hypothesis is formulated:

H7. Project coordination is positively related to project performance.

3.4 Effects of project commitment

Project commitment, as a form of emergent state (i.e. cognitive, motivational and affective states; Marks *et al.*, 2001), has been identified as a good predictor of project performance across various industries, including the automotive industry, new product development projects and construction projects (Buvik and Tvedt, 2017; Ehrhardt *et al.*, 2014; Hoegl *et al.*, 2004). In construction projects, there commonly exist numerous teams in a project. These teams have a natural tendency to focus on their own team goals (Hoegl *et al.*, 2004). Acceptance of project goals and values enables project teams to focus not only on their own benefits but also on the overall performance of the project. Besides, in a project with low commitment, project teams tend to ignore the requirements of other teams and not improve project performance even if they can do it easily, such as informing other teams of safety risks, which will be detrimental to project performance. Thus, we formulate the following hypothesis:

H8. Project commitment is positively related to project performance.

4. Methods

4.1 Measurement development

Multi-item measurement scales derived from the existing literature were used to develop a structured questionnaire. Besides, we employed the backward translation approach to ensure consistency between the Chinese and original English versions of measurement items with the assistance of two professors and three doctoral candidates in project management. Specifically, the dependent variable, IMTBS, was measured by the items from [Faraj and Yan \(2009\)](#) and [Lee and Sawang \(2016\)](#). The scale used to measure knowledge integration (KI) was based on the research of [Mehta and Mehta \(2018\)](#). The scale for project coordination (PCO) was adapted from [Hoegl et al. \(2004\)](#) and [Siebdrat et al. \(2014\)](#). The project commitment (PCM) items were designed with reference to the research of [Hoegl et al. \(2004\)](#). Finally, a five-item instrument was used to measure project performance (PP; [Lu et al., 2019](#)).

Furthermore, online interviews with experts in construction project management were conducted to ensure the appropriateness, intelligibility and sensitivity of these items. A total of seven experts from five companies, including project manager, department manager and project engineer, participated in the interview. To resolve disagreements among these experts, we sent the revised version of the survey items to each expert to achieve their approval. After three-round revisions, they reach a consensus on the final version of the survey items, which are listed in [Table A1](#). Each item was measured by a seven-point Likert scale.

4.2 Data collection

Data were collected from construction projects in China. We first conducted a pilot study involving 30 respondents from the Master of Engineering Management (MEM) students majoring in construction management at a Chinese renowned university to test the validity of all constructs. All of them are full-time employees in the construction industry. They are all part-time students and commonly participate in the MEM courses on the weekend ([Jia et al., 2021a](#); [Ma et al., 2021](#)). The validity of all constructs is satisfactory in the pilot study. Then, the questionnaire was distributed to construction professionals to report the condition of the last completed project they managed. We randomly sent the link to the online survey to alumni of two renowned universities, who were involved in the construction industry. A total of 400 questionnaires were distributed to these alumni from 53 construction companies, and 256 responses were returned. After removing invalid responses, 206 valid questionnaires were left, thereby yielding a valid response rate of 51.5%. The 50 invalid questionnaires resulted from incomplete data and short answer time (less than 1 minute). The descriptive information of the sample respondents and projects is given in [Table 1](#).

According to [Table 1](#), most respondents are male (67%), which corresponds to the characteristic of the construction industry. A total of 88% of the informants have received an undergraduate degree or above. Further, 15% of the sampled projects last less than or equal to 6 months, 20% last 7–12 months, 15% last 13–18 months, 12% last 19–24 months and 38% last more than 24 months. Regarding project cost, the total investment cost of 69 projects is less than or equal to 50 million RMB (approximately USD 7.5 million), of 35 projects is more than 50 million but less than 100 million RMB (approximately USD 15 million), 65 projects is more than 100 million but less than 1 billion RMB (approximately USD 150 million) and of 37 projects is more than 1 billion RMB, corresponding to the rates of 33%, 17%, 32% and 18%, respectively. Finally, about 43% of the sampled projects are building construction, 28% are infrastructure construction and the rest 29% belong to other types of construction.

This study employs two tests to eliminate potential common method bias. First, Harman's single-factor method is used to test the data by estimating a model in which all the model indicators are loaded on a single factor ([Podsakoff et al., 2003](#)). The results show no serious common method bias with a bad fit with the data ($\chi^2(170) = 991.2, p < 0.001$, CFI = 0.727,

Table 1.
Descriptive
information of
respondents and
projects (*N* = 206)

Measure	Item	Frequency	Percentage
Gender	Female	69	33
	Male	137	67
Education	Junior college or below	25	12
	Undergraduate	137	67
	Master's or above	44	21
	6 months or below	32	15
Project duration	7–12 months	41	20
	13–18 months	30	15
	19–24 months	24	12
	25 months or above	79	38
	50 million or below	69	33
Project cost (RMB)	51–100 million	35	17
	101–1000 million	65	32
	1001 million or above	37	18
Project type	Building construction	89	43
	Infrastructural construction	57	28
	Others	60	29

TLI = 0.695, RMSEA = 0.154). Second, the common method factor approach suggested by Podsakoff *et al.* (2003) is employed to examine the data. Compared to the baseline model (five-factor model), the addition of the common method factor does not improve the fit of the model greatly (CFI: 0.939 vs 0.963, TLI: 0.927 vs 0.950, RMSEA: 0.075 vs 0.062). In this way, common method bias is not a significant issue in the data.

5. Results

A covariance-based structural equation modeling (CB-SEM) technique is used to analyze the data. In this study, CB-SEM is preferable as this study aims to test theory. AMOS, as a basic tool for CB-SEM analyses, is employed in the current study.

5.1 Measurement model

We first analyze the measurement model to test the validity and reliability of constructs. As shown in Table 2, the standardized factor loadings of all items are slightly less than or greater than the acceptable level of 0.7, indicating the items in this study are reliable (Chin, 1998). Then, Cronbach's α and composite reliability are calculated to assess the reliability (Fornell and Larcker, 1981). The constructs' Cronbach's α ranges from 0.795 to 0.904, and composite reliability ranges from 0.798 to 0.906, all exceeding the 0.7 thresholds. Furthermore, values of the average variance extracted (AVE) ranging from 0.569 to 0.706 are above 0.50 (Flynn *et al.*, 1990). Finally, to test the discriminant validity, models combining different constructs and the baseline model (including IMTBS, KI, PCO, PCM and PP) are compared. As shown in Table 3, the goodness-of-fit (GOF) of the baseline model is better than other models, thereby establishing good discriminant validity. We also examine variance inflation factors (VIFs) to test for potential multicollinearity. The analysis results suggest that multicollinearity is not a significant problem as the highest value of VIF (4.17) is below 5 (Hair *et al.*, 2016).

5.2 Structural model

The GOF measures recommended by Chau (1997) are employed to evaluate the structural model. The ratio of chi-square and degree of freedom is 2.13, which is lower than the acceptable level of 3 suggested by Chau (1997). The Tucker–Lewis index (TLI) and

						From local to the whole
Constructs	Items	Loadings	Cronbach's alpha	Composite reliability	Average variance extracted	
IMTBS	IMTBS1	0.810	0.904	0.906	0.706	363
	IMTBS2	0.861				
	IMTBS3	0.874				
	IMTBS4	0.814				
KI	KI1	0.803	0.879	0.880	0.648	
	KI2	0.849				
	KI3	0.793				
	KI4	0.772				
PCO	PCO1	0.797	0.795	0.798	0.569	
	PCO2	0.718				
	PCO3	0.746				
PCM	PCM1	0.699	0.894	0.898	0.690	
	PCM2	0.881				
	PCM3	0.836				
	PCM4	0.892				
PP	PP1	0.825	0.899	0.901	0.646	
	PP2	0.794				
	PP3	0.775				
	PP4	0.813				
	PP5	0.811				
						Table 2. Results of reliability and validity test

Table 2.
Results of reliability and validity test

Model	χ^2	df	χ^2/df	CFI	TLI	RMR	RMSEA
Baseline model	344.3	160	2.2	0.939	0.927	0.072	0.075
Single-factor model	991.2	170	5.8	0.727	0.695	0.191	0.154
Two-factor model	867.3	169	5.1	0.768	0.739	0.170	0.142
Three-factor model 1	758.1	167	4.5	0.803	0.730	0.168	0.131
Three-factor model 2	490.8	167	2.9	0.892	0.877	0.095	0.097
Four-factor model 1	722.8	164	4.4	0.814	0.785	0.162	0.129
Four-factor model 2	449.1	164	2.7	0.905	0.890	0.090	0.092
Four-factor model 3	441.4	164	2.7	0.908	0.893	0.084	0.091

Note(s): Baseline model: IMTBS, KI, PCO, PCM, PP; single-factor model: IMTBS + KI + PCO + PCM + PP; two-factor model: IMTBS + KI, PCO + PCM + PP; three-factor model 1: IMTBS + KI + PCO, PCM, PP; three-factor model 2: IMTBS, KI + PCO + PCM, PP; four-factor model 1: IMTBS + KI, PCO, PCM, PP; four-factor model 2: IMTBS, KI + PCM, PCO, PP; four-factor model 3: IMTBS, KI, PCO, PCM + PP

Table 3.
Results of discriminant validity test

comparative fit index (CFI) values are 0.928 and 0.939, respectively. They are all above the acceptable level of 0.90. Further, the root mean square error of approximation (RMSEA) value is 0.074, which is below the recommended level of 0.1 (Chau, 1997). Thus, a good fit between the data and the proposed model is suggested.

The results of hypothesis testing are summarized in Table 4. Hypotheses 1, 2 and 3 predict a direct effect of IMTBS on knowledge integration, project coordination and project commitment. The results suggest direct positive effects for IMTBS on project coordination ($\beta = 0.14, p < 0.05$) and knowledge integration ($\beta = 0.51, p < 0.001$), but a negative effect for IMTBS on project commitment ($\beta = -0.13, p < 0.05$). Therefore, Hypotheses 1 and 2 are supported, whereas Hypothesis 3 is not supported. The influence of knowledge integration on project coordination is statistically significant ($\beta = 0.78, p < 0.001$), and the influence of knowledge integration on project commitment is also statistically significant ($\beta = 0.60, p < 0.001$). Thus, Hypotheses 4 and 5 are supported. In addition, the direct effects of project

coordination on project commitment ($\beta = 0.35, p < 0.05$) and project performance ($\beta = 0.42, p < 0.001$) are both statistically significant, thereby approving [Hypotheses 6](#) and [7](#). Finally, project commitment is found to have a significant positive relation with construction project performance ($\beta = 0.52, p < 0.001$), which supports [Hypothesis 8](#).

5.3 Mediation analysis

The results of the mediation analysis are exhibited in [Table 5](#). We employ the bootstrapping technique (random sampling with 5,000 resamples) to analyze the mediating effects among the constructs. As shown in [Table 5](#), project coordination cannot mediate the relationship between IMTBS and project performance, while project commitment negatively mediates such a relationship. Besides, knowledge integration is found to mediate the link between IMTBS and project coordination as well as the link between IMTBS and project commitment. Further, knowledge integration and project coordination are found to make a serial mediating effect on the relationship between IMTBS and project performance, and so are knowledge integration and project commitment for this relationship. In addition, the relationship between IMTBS and project commitment is mediated by knowledge integration but not project coordination. Finally, after controlling the effect of IMTBS on project performance, we perform the mediation analysis again. The significant level of coefficients does not change, while the direct effect of IMTBS on project performance is not significant, thus further proving the multiple mediating effects ([Khosravi et al., 2020](#)).

6. Discussion

6.1 Main findings

From a MTS perspective, combining the IPO framework and coordination theory, this study conducts a conceptual model to analyze how IMTBS influences construction project performance.

Table 4.
Hypothesis testing
result

Hypothesis	Result
H1 : IMTBS is positively related to knowledge integration	Confirmed
H2 : IMTBS is positively related to project coordination	Confirmed
H3 : IMTBS is positively related to project commitment	Not confirmed
H4 : Knowledge integration is positively related to project coordination	Confirmed
H5 : Knowledge integration is positively related to project commitment	Confirmed
H6 : Project coordination is positively related to project commitment	Confirmed
H7 : Project coordination is positively related to project performance	Confirmed
H8 : Project commitment is positively related to project performance	Confirmed

Table 5.
Mediating effect test

Path	Effect	<i>p</i>	95% bootstrapping confidence interval	
			Lower bound	Upper bound
IMTBS → PCO → PP	0.051	0.054	−0.001	0.182
IMTBS → PCM → PP	−0.058	0.035	−0.166	−0.004
IMTBS → PCO → PCM	0.040	0.080	−0.002	0.172
IMTBS → KI → PCM	0.262	0.009	0.095	0.511
KI → PCO → PP	0.334	0.002	0.123	0.642
KI → PCM → PP	0.320	0.022	0.055	0.613
IMTBS → KI → PCO → PP	0.147	0.001	0.056	0.300
IMTBS → KI → PCM → PP	0.141	0.016	0.030	0.325

Data collected from 206 Chinese construction projects are used to empirically verify the proposed research model. In general, most of the proposed hypotheses receive empirical support except [Hypothesis 3](#). Mediation analysis is also conducted to further reveal the link between IMTBS and project performance. Then we will discuss these findings.

Contrary to our formulation, IMTBS has a significant negative instead of a positive influence on project commitment. Based on previous research ([Bolton et al., 2021](#)), we proposed that frequent and timely communication is the foundation of relational coordination. Although instant messaging facilitates frequent and timely boundary spanning, we ignored its potential drawbacks, which potentially explain this negative relationship. IMTBS removes time and space limits and thus enables instant information flows ([Huang et al., 2020](#)). Accompanying these benefits, IMTBS also brings about potential interruptions for other teams ([Gupta et al., 2009](#); [Sun et al., 2021](#)). Other teams have to spare time, energy and attention to deal with information or resource requirements from the local team. The component teams seem to be around the clock to provide feedback to other teams ([Gupta et al., 2009](#)). From an information flow perspective, IMTBS promotes information exchange among component teams, thus helpful for task completion, project coordination and knowledge integration. Yet, from an affective perspective, it is frustrating to be constantly interrupted by other teams' requirements, especially in an unintended way ([Van Osch and Steinfield, 2016](#)). As such, IMTBS deteriorates project commitment.

Although IMTBS is verified to positively influence both project coordination and knowledge integration, the strength between IMTBS and knowledge integration is much larger than that between IMTBS and project coordination. In construction projects, the component team is the agency of its parent organization and occupies specific cooperation tasks. Yet, before cooperation, a contract commonly has been reached between their parent organizations, which greatly determines project coordination ([Ma et al., 2020](#); [You et al., 2019](#)). While the component teams are responsible for performing the project, they have to obey the contract. In this way, the influence of IMTBS (team-level) on project coordination is limited. However, component teams play a critical role in coordinating their respective knowledge because knowledge is embedded in teams instead of their parent organization. As a result, the effect of IMTBS on knowledge integration is relatively large.

While previous research suggests a small effect of knowledge sharing on project coordination in construction projects ([Ma et al., 2021](#); [Wen and Qiang, 2016](#)), knowledge integration is found to be a strong contributor to project coordination in this study. The possible explanation is that knowledge integration requires recognition of other teams' knowledge and understanding of knowledge structures. As a result, teams deepen their mutual understanding of others' task requirements and the whole project operation, thus greatly promoting project coordination. What is more, project coordination is found to positively influence project performance, which is consistent with prior research ([Hoegl et al., 2004](#)). Project commitment is also found to have a significant positive relationship with project performance, which is consistent with previous research on other types of projects ([Ehrhardt et al., 2014](#); [Hoegl et al., 2004](#)). In this study, project commitment is found to be a powerful predictor of project performance, which echoes the emphasis on the importance of emergent states in MTSs ([Kou, 2021](#); [Zaccaro et al., 2020](#)).

This study verifies the mediating role of knowledge integration in the link of IMTBS and project coordination, IMTBS and project commitment. Furthermore, knowledge integration and project coordination, knowledge integration and project commitment are confirmed to make serial mediating effects on the relationship between IMTBS and construction project performance. These findings affirm the important role of knowledge integration in leveraging the individual team's endeavor to construction MTS functioning. In practice, knowledge integration helps decrease knowledge differences among teams and develop common knowledge within projects ([Majchrzak et al., 2012](#)). In this way, teams could have a better understanding of others' tasks and

how the project should operate as a whole. These arguments provide a possible explanation for the significance of knowledge integration in construction projects. In addition, the mediating role of project coordination between IMTBS and project performance is not supported, while project commitment is found to mediate such a relationship. A possible explanation is that, as the executive subjects of project tasks, component teams could span their boundary to influence project coordination. However, such influence is limited and could not be leveraged for improving project performance. On the contrary, as IMTBS hampers the strong belief in the goals of the project as well as the willingness to engage in the project, teams in the project tend to focus more on their own goals instead of the collective project goals, which is detrimental to the construction project performance (Luciano *et al.*, 2018). The above findings suggest that a component team's boundary spanning may not necessarily contribute to, but could damage, the overall performance.

6.2 Theoretical implications

This study contributes to the literature in the following aspects.

First, this study contributes to construction management literature. Although previous research has paid attention to boundary-spanning behavior in construction projects (Di Marco *et al.*, 2010; Lee and Sawang, 2016; Sackey and Akotia, 2017; Stjerne *et al.*, 2019), most of them focused on boundary-spanning behavior itself or its antecedents. Little attention has been paid to the consequence of boundary spanning. Besides, previously studied boundary spanning is limited to the organizational or the individual level. The project team is the agency of their parent organizations and the subject of construction performance, yet team-level boundary spanning has been neglected. This study sheds light on team-level boundary spanning and explores its effects on system-level or project performance, thus filling the identified literature gap in the construction field.

Second, with coordination theory, this study unravels the underlying mechanism between IMTBS and construction project performance. IMTBS is a kind of interfacing activity and becomes increasingly popular in today's digital and post-pandemic era. It inevitably affects inter-team or system-level coordination. Yet, how IMTBS influences different forms of coordination and their differential mediating effects has not been comprehensively investigated in the literature (Bienefeld and Grote, 2014; Davison *et al.*, 2012; Kou, 2021). From task, knowledge and relational perspectives, three forms of coordination in construction projects are explored simultaneously. Their differential mediating effects are also revealed, thus contributing to the literature on team boundary spanning in the construction field.

Third, this study contributes to the MTS literature. In the MTS literature, the effect of TBS on system performance has been investigated but does not reach consistency (Davison *et al.*, 2012; Gopal and Gosain, 2010; Hoegl *et al.*, 2004). As previous scholars suggested (Luciano *et al.*, 2018; Zaccaro *et al.*, 2020), we focused on a specific form of MTS, that is, construction projects, typical large-scale MTSS in practice. Besides, both the positive and negative effects of IMTBS are found in this study, which helps to explain the mixed results in the literature. Furthermore, previous research argued that the effect of TBS on system performance depends greatly on the research context, that is specific forms of MTS (Luciano *et al.*, 2018; Zaccaro *et al.*, 2020). However, results in this study indicate that the form of TBS is also a significant contributor, which should attract more attention in the future. In this way, these findings enrich our understanding of the effects of TBS in MTS and thus contribute to the MTS literature.

6.3 Practical implications

This study provides several implications for construction management practice. First, managers should recognize the double-edged characteristic of IMTBS. Although IMTBS is

beneficial to task and knowledge coordination, it also hampers project commitment. In this way, we suggest two possible solutions. On the one hand, regulations or formal communication protocols on the use of instant messaging could be formulated, so as to avoid the unintended penetration and interruption of IM into other teams' operations (Ma *et al.*, 2021). On the other hand, during the process of IMTBS, the component team should focus not only on information exchange and task coordination, but also on the change of other teams' emergent states (e.g. project commitment). For instance, they should try to avoid expressing in a confusing way and communicate in a moderate way by instant messaging. Second, the mediation analysis indicates that knowledge integration is an important bridge that links IMTBS and construction project performance. As such, construction managers are recommended to cultivate a learning-oriented climate to encourage different opinions and ideas to surface, thereby facilitating project learning (Mu *et al.*, 2021). Finally, measures should also be taken to improve project commitment. For example, some construction projects, such as public projects, are meaningful to local residents or citizen well-being, which should be stressed by managers such that each team values being part of this project (Green and Sergeeva, 2019).

7. Conclusion

In recent years, the COVID-19 pandemic accelerated the adoption of instant messaging, making IMTBS prevalent in construction projects. Yet, the influence of IMTBS as well as the underlying mechanisms is still blurry. Combining the MTS perspective and coordination theory, this study finds three forms of coordination as critical mediators that link IMTBS to construction project performance. Of note, project commitment negatively mediates the relationship between IMTBS and project performance. In summary, the proposed model affords a good explanation of the influence of IMTBS on construction project performance.

Although this study provides valuable implications, there also inevitably exist several limitations. One limitation is that this study employs cross-sectional data to verify the proposed model. Future research should conduct a longitudinal design survey to explore the effects of IMTBS over time as IMTBS influences not only the current but also the future project processes and intermediate outcomes. What is more, although this study examines the influence of IMTBS on construction projects, the interplay of different forms of TBS (e.g. face-to-face, instant messaging, building information modeling) requires further investigation as the interactions between component teams could occur both physically and virtually. In addition, the sample in this study is from a single country. Future research should generalize the findings with data from other countries.

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Further reading

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Construct	Item	Measure
Instant messaging-based team boundary spanning	IMTBS1	My team often solicits information and resources from other teams by instant messaging (e.g. WeChat and QQ)
	IMTBS2	My team tries to influence important actors in other teams by instant messaging (e.g. WeChat and QQ)
	IMTBS3	My team keeps other teams in the project informed of my team's activities by instant messaging (e.g. WeChat and QQ)
	IMTBS4	My team proactively seeks the advice and support of other teams by instant messaging (e.g. WeChat and QQ)
Knowledge integration	KI1	Each team pools in their expertise to jointly solve project-related problems
	KI2	Many creative ideas and schemes come from the discussions among all teams
	KI3	Teams in our project frequently build on each other's ideas, skills and expertise to develop new project-related knowledge for decision-making
	KI4	Teams in our project often gained new insights by sharing their ideas with each other
Project coordination	PCO1	The work done by each team is closely harmonized in our project
	PCO2	Our project establishes ground rules, routines and meeting schedules to facilitate project work
Project commitment	PCO3	Each team in our project avoids duplication of effort
	PCM1	Each team feels fully responsible for achieving the common project goals
	PCM2	This project has the strong commitment of all teams
	PCM3	Each team is committed not only to their teams, but to the overall project
Project performance	PCM4	Each team values being part of this project
	PP1	The project results, or deliverables, are in line with the client objectives
	PP2	This project is within the budget
	PP3	This project is on schedule
	PP4	The construction and deliverables quality accord with the standard
	PP5	The participants of this project maintain good cooperation

Table A1.
Measurement items

Corresponding author

Guofeng Ma can be contacted at: 06125@tongji.edu.cn

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